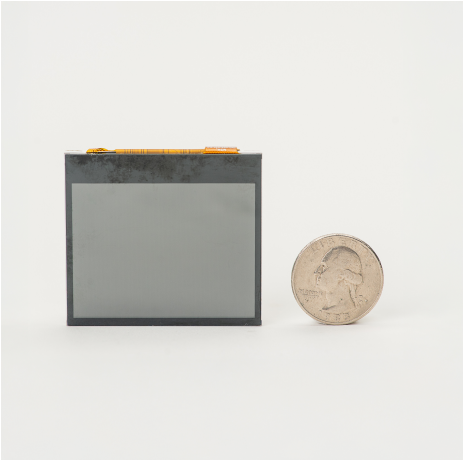
OpenMV® Touch LCD Shield

Datasheet

SKU: OMV-TOUCH-LCD-SHIELD-V3

Description

The Touch LCD Shield gives your OpenMV Cam the ability to display what it sees on the go while not connected to your computer. It features a 2.3-inch 320x240 16-bpp (RGB565) capacitive multi-touch TFT LCD display with a controllable backlight. The shield also includes dual QWIIC headers connected to P4 (SCL) and P5 (SDA) for daisy-chaining I2C peripherals from SparkFun's QWIIC ecosystem. Controlled via the SPI Display Class.



Target Areas

Touch interface, portable display, interactive demos, HMI, field debugging, sensor integration via QWIIC.

Key Features

- **2.3" touch display** — 320x240 pixel, 16-bit color (RGB565), capacitive multi-touch.
- **Controllable backlight** — Software-controllable backlight brightness.
- **Dual QWIIC** — Two QWIIC I2C connectors on P4/P5 for daisy-chaining SparkFun QWIIC peripherals.
- **Built-in firmware support** — Controlled via the SPI Display Class in MicroPython.

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1 Features

1.1 Display

Component	Details
Screen	2.3-inch TFT LCD, 320x240 pixels, 16-bit color (RGB565).
Touch	Capacitive multi-touch input.
Backlight	Software-controllable backlight brightness.
Interface	SPI connection to OpenMV Cam. Controlled via the SPI Display Class.

1.2 Connectors

Component	Details
QWIIC	Dual QWIIC I2C connectors. Connected to P4 (SCL) and P5 (SDA). Compatible with SparkFun QWIIC ecosystem for daisy-chaining I2C peripherals.

1.3 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.

2 General Info

Dimension	Value
Weight	25 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/touch-lcd-shield
SPI Display Class	https://docs.openmv.io/library/omv.display.spidisplay.html
SparkFun QWIIC	https://www.sparkfun.com/qwiic
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.